Quantum Secure Cryptographic Library (QSC)

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1) Overview

The Quantum Secure Cryptographic (QSC) Library, version 1.4.0.0, is QRCS Corporation's common cryptographic software foundation. Written in C using a MISRA-adjacent development discipline, it brings together classical and post-quantum cryptography, hashing, authentication, key derivation, TLS 1.3, X.509 PKI, networking, and secure systems utilities. The library now couples these capabilities with configurable, size-conscious CMake compilation and optional hardware-backed private-key interfaces. It supports a spectrum of deployments, from constrained devices and industrial gateways to enterprise services, without requiring separate cryptographic stacks for each product. QRCS-developed constructions, including RCS, CSX-512, QMAC, and SCB, complement its standards-oriented primitives.

2) Motivation and Strategic Rationale

The transition to quantum-resistant public-key cryptography creates a practical challenge: organizations must modernize algorithms, credentials, and protocols while maintaining interoperability with existing systems. QSC addresses that integration burden through a reusable implementation platform that combines standardized post-quantum KEMs and signatures with established classical algorithms, authenticated encryption, certificate management, and system-level utilities. A shared codebase enables QRCS products and third-party adopters to reuse components, consolidate review effort, and plan phased migrations. Its strategic value lies in technical breadth, integration flexibility, and the associated source, API, and test assets rather than in any single primitive.

3) Architecture and Mechanism

Symmetric ciphers: QSC includes two QRCS-developed authenticated-encryption constructions: CSX-512, a ChaCha-derived design with a 512-bit key and expanded state, and RCS, a Rijndael-based design with cSHAKE-derived key material and KMAC authentication. Conventional implementations include AES-128/256 with CBC, CTR, ECB, GCM, and HBA interfaces, together with ChaCha20-Poly1305. Authenticated modes are preferred for application data; callers using unauthenticated modes must enforce nonce, IV, padding, and integrity requirements in their encompassing protocols.

Hashing, MAC, and KDF: SHA-2 and the Keccak family underpin QSC's hashing, extensible-output, MAC, and derivation services. The library provides SHA-3, SHAKE, cSHAKE, KMAC, HMAC, and HKDF, with one-shot and state-based interfaces and selected accelerated implementations. QMAC supplies a QRCS GF(2^256)-based authentication construction, while SCB provides a configurable SHAKE-based, cost-oriented derivation mechanism. The common API design supports reuse across secure channels, credential systems, and device provisioning.

Asymmetric and PQC: Classical interoperability is provided by NIST P-256, P-384, and P-521 ECDH/ECDSA, together with X25519/X448 key agreement and Ed25519/Ed448 signatures. The standardized post-quantum portfolio includes ML-KEM (FIPS 203), ML-DSA (FIPS 204), and SHAKE-based SLH-DSA profiles (FIPS 205). Additional Classic McEliece, HQC, Falcon, and research-oriented parameter options expand the library's algorithm-selection and interoperability capabilities. Standardized, candidate, and QRCS-developed mechanisms remain distinct within the library's interface and documentation model.

Randomness and KDF support: Entropy and random-generation facilities comprise ACP, CSP, RDP, CSG, HCG, and SECRAND. Revision 1.4 adds the External True Random (ETR) interface, which accepts an external entropy source and applies input validation and SHAKE-512 conditioning. Together, these facilities support cryptographic key generation, protocol initialization, device provisioning, and reproducible test workflows, with provider selection governed by application and platform policy.

Systems framework: QSC integrates TLS 1.3 clients, servers, record protection, handshake and transcript processing, certificate authentication, named-group negotiation, and socket adapters. Its X.509 PKI toolkit covers certificate and CSR creation, chain verification, CRLs, OCSP, PKCS#12, PEM, and trust stores. Supporting modules provide IPv4/IPv6 networking, asynchronous execution, thread pools, ASN.1/BER/DER encoding, data structures, file services, CPU feature detection, and portable utility functions. These components make QSC a systems-integration library as well as a catalog of cryptographic algorithms.

Hardware security interfaces: Revision 1.4 introduces a unified provider contract and optional adapters for PKCS#11 HSMs, Windows NCrypt/TPM, Linux TPM2-TSS, Arm PSA Crypto, selected secure elements, and platform keystores. TLS/X.509 signing bridges integrate externally managed private keys into certificate and secure-channel workflows. The provider architecture supports backend-specific authorization, key custody, capability discovery, and attestation policies.

Secure memory: Dedicated utilities provide controlled allocation, movement, comparison, clearing, and explicit erasure, with selected platform-aware and timing-resistant implementations. These services give applications consistent tools for secret-lifetime management, controlled memory handling, and integration with software and protected-key backends.

4) Security Model and Post-Quantum Posture

QSC is designed for applications transitioning from classical to post-quantum cryptography. Its security architecture combines authenticated encryption, standards-aligned public-key mechanisms, cryptographic hashing and derivation, explicit memory handling, and configurable build profiles. TLS 1.3 and X.509 support certificate-based secure channels, while hardware-provider interfaces permit selected private-key operations to be delegated to configured external services. The library's design emphasizes explicit algorithm selection, traceable interfaces, and reproducible evaluation at the deployment level.

The project maintains cryptographic known-answer and regression tests, CAVP/ACVP-oriented vector assets, protocol and certificate test suites, benchmarks, and platform-specific build workflows. These artifacts support internal engineering qualification, customer evaluation, and independent technical assessment across portable, accelerated, and hardware-provider configurations. Formal FIPS 140-3 module validation, ACVP/CAVP algorithm validation, and independent side-channel certification are separate external processes; none is claimed for QSC 1.4.0.0 in this summary.

5) Implementation and Integration

QSC provides a consistent C API with state-oriented interfaces for native static-library integration, developer tooling, and reuse throughout QRCS applications. Visual Studio, GCC/Clang, and CMake build paths support cross-platform development, while QSCTest, conformance utilities, and managed wrapper assets extend the integration and evaluation environment. The technical specification and generated reference material describe the

application-facing APIs, with implementation-specific routines maintained as separate lower-level interfaces.

Optimized compilation: The modular CMake integration defines seven deployment profiles: FULL, CLASSICAL, PQC3, PQC5, HYBRID, DEVICE_IDENTITY, and BOOT_VERIFY. Each profile selects source components and feature definitions for its deployment objective, enabling smaller, more focused distributions for constrained devices, protocol stacks, enterprise services, and boot or identity applications. The profile architecture supports repeatable build configuration and product-specific packaging.

Developers can integrate secure-channel functionality through the higher-level TLS socket and certificate interfaces, leaving record framing, transcript management, key scheduling, and extension processing to the protocol layer. For PKI workflows, the X.509 wrappers, certificate/CSR builders, verification, trust-store, PEM, OCSP, and PKCS#12 interfaces provide practical entry points. The underlying networking, asynchronous, and event-loop services are available for applications that require greater control over transport and concurrency.

Performance-sensitive implementations use AES-NI, supported SHA instructions, AVX2/AVX-512, and other architecture-specific paths when available, with portable implementations supporting additional targets. Configurable compilation, CPU capability detection, and the benchmarking framework provide a foundation for balancing throughput, memory footprint, binary size, and portability across deployment classes.

6) Use Cases and Applications

Embedded and regulated systems: A configurable static-library footprint, explicit interfaces, and secure-memory facilities make QSC relevant to embedded controllers, industrial gateways, device identities, and boot verification. Modular compilation permits a device vendor to select the required primitives and supporting components without carrying unrelated protocols, reducing integration complexity in memory- and storage-constrained systems.

Secure communications stacks: QSC unifies classical and post-quantum key establishment, authenticated encryption, TLS 1.3, X.509 credentials, and networking services for secure gateways, VPNs, messaging, service authentication, and QRCS communications protocols. The TLS implementation provides certificate authentication, hybrid groups, ALPN, SNI, mutual TLS, session tickets, and protected application records, creating an integrated foundation for secure-channel products.

Enterprise key management and PKI services: QSC's X.509 toolkit, ML-DSA and classical signing APIs, Keccak and SHA-2 derivation primitives, and certificate verification services support

enterprise PKI, trust administration, signing, and key lifecycle management. Optional PKCS#11 and operating-system key providers connect the software to external key custody and authorization infrastructure, allowing organizations to align cryptographic operations with their existing security architecture.

Critical infrastructure and ICS/SCADA: Long-lived industrial and critical-infrastructure deployments benefit from staged software modernization, controlled configuration changes, and predictable implementation footprints. QSC combines post-quantum algorithms, tailored compilation, TLS/X.509 services, and portable C integration to support phased cryptographic migration across field-upgradable systems and equipment generations.

Research and national security programs: The breadth of cryptographic implementations, reference vectors, benchmark harnesses, and protocol integration interfaces makes QSC a foundation for research, laboratory evaluation, and sovereign cryptographic programs. Accessible implementation code and test assets support reproducible technical studies, algorithm comparisons, and cooperative assessment across academic, commercial, and government environments.

7) Economic and Operational Value

Integration velocity and maintainability: A unified cryptographic API, shared memory and platform utilities, and reusable TLS/X.509 services enable teams to build multiple products from one maintained implementation foundation. Profile-based source selection gives integrators a practical way to align cryptographic functionality with deployment requirements, reducing duplicated engineering and simplifying long-term maintenance.

Assurance and diligence value: QSC combines inspectable source code, an organized test harness, standards-oriented vectors, and detailed API references in a single diligence package. This provides clients, technical partners, and prospective acquirers with tangible materials for examining algorithm implementations, integration boundaries, platform behavior, and the library's maintainability.

Performance and deployment efficiency: Optimized compilation and hardware-accelerated code paths give implementors flexibility to balance application footprint, throughput, and portability. The benchmarking framework and profile-based configurations allow performance and deployment economics to be assessed using representative workloads and target hardware, supporting informed engineering and procurement decisions.

Portfolio leverage: As the shared cryptographic foundation for QRCS products, QSC creates leverage across secure gateways, communications systems, key-management services, and

developer SDK offerings. Reusable primitives, certificate tooling, protected-key adapters, and cross-platform infrastructure make the library a coherent intellectual-property asset with applications across multiple commercial product categories.

8) Long-Term Security Benefit

QSC brings standardized post-quantum mechanisms together with interoperable classical algorithms, authenticated encryption, TLS 1.3, X.509 credentials, and deliberate secret-management facilities. This combination supports staged protection of long-lived information while preserving compatibility with existing networks and credential infrastructure. Its modular architecture also enables continuing algorithm evolution without replacing the entire systems framework.

The library's standards-oriented components include AES, SHA-2, SHA-3, SHAKE, KMAC, TLS 1.3, X.509, ML-KEM, ML-DSA, and SLH-DSA. Candidate algorithms and QRCS-developed constructions are identified separately, giving implementors a clear basis for evaluating interoperable profiles alongside specialized cryptographic capabilities.

A consistent C API, controlled memory interfaces, modular CMake builds, reproducible test assets, and optional protected-key integrations support maintainable deployment across embedded, industrial, enterprise, and public-sector environments. The result is a practical cryptographic foundation that connects algorithm modernization with secure channels, certificates, identity, and key management.

9) Conclusion

QSC 1.4.0.0 is a strategically reusable QRCS technology asset, combining classical and post-quantum cryptography, QRCS-developed authenticated-encryption constructions, TLS 1.3, X.509 PKI, optional hardware-backed key interfaces, and configurable deployment profiles in a common C library. For customers, it offers a cohesive platform for secure communications, digital identity, signing services, and cryptographic modernization. For partners, investors, and prospective acquirers, it represents a substantial body of reusable intellectual property, engineering infrastructure, and integration capability with applications across enterprise, embedded, industrial, and sovereign-security markets.

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